



實習醫學生臨床技能核心課程

6.3 肌肉骨骼磁共振造影判讀

台北醫學大學 影像醫學部

教學目標

- 說明各項基礎磁共振造影檢查的適應症及禁忌(包括顯影劑的安全使用)。
- 具備基礎磁共振造影學及相關部位解剖學知識。
- 確認基礎磁共振造影影像病人姓名、檢查日期及影像方向(orientation)。
- 系統性的描述基礎磁共振造影影像，並指出病灶之型態及特性。
- 判讀常見的異常基礎磁共振造影影像，並且列出鑑別診斷。

教學活動

- 基本能力: 說明磁共振造影的適應症及禁忌；
確認磁共振造影影像的基本資料；基礎磁共振造影及骨骼，軟骨，關節及軟組織解剖學知識
- 肌腱及韌帶損傷
- 骨骼，軟骨及關節疾病
- 軟組織腫瘤：脂肪瘤，神經瘤
- 測驗

磁共振造影檢查的適應症

- 沒有游離輻射的困擾
- 利用許多不同的脈衝序列(sequence)、多方位的造影技術，對於骨髓疾病及軟組織結構的影像評估特別良好
- 病患可不需注射對比劑即可評估組織結構

磁共振造影檢查的禁忌

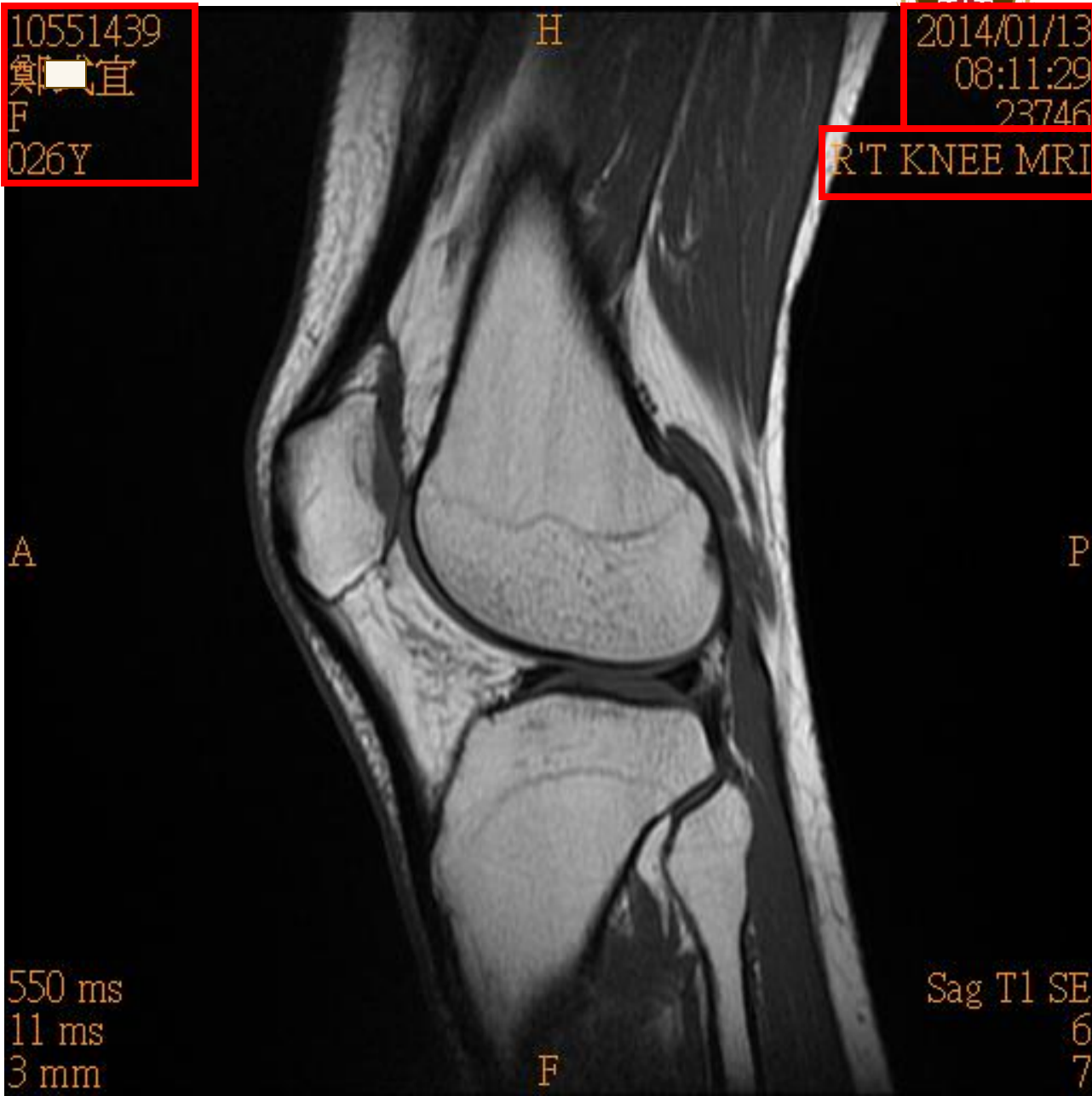
- 非固定式體內金屬植入物 (如心律調節器、人工心臟瓣膜、血管夾、電子耳蝸等)
- 幽閉恐懼症
- 藥物過敏
- 腎功能不全, 可能造成 Nephrogenic systemic fibrosis (NSF)

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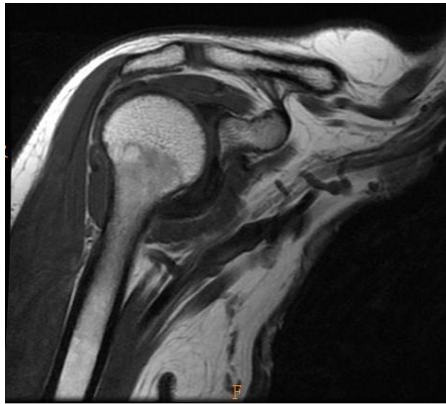
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R'T KNEE MRI

確認病人
姓名、檢
查日期及
影像方向
(orientation)

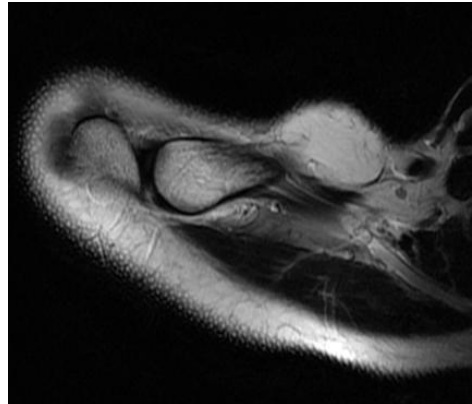


磁共振影基礎序列 (sequence): 提供不同軟組織對比



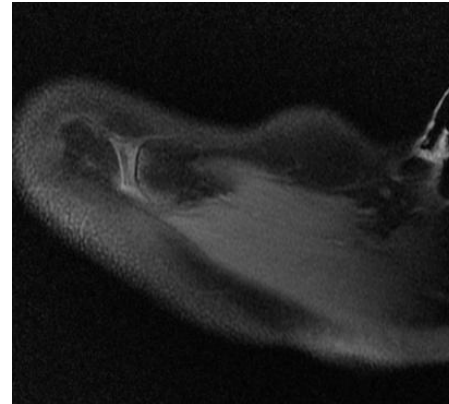
T1-weighted image

T1 權重影像



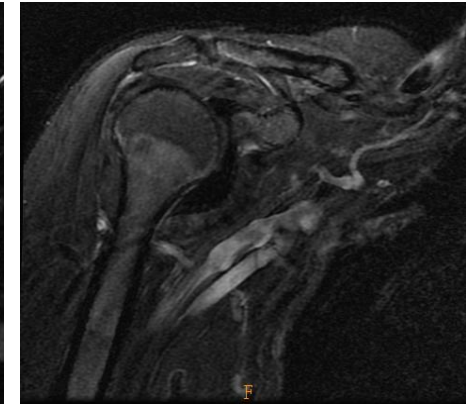
T2-weighted image

T2權重影像



**T1-weighted
with fat suppression**

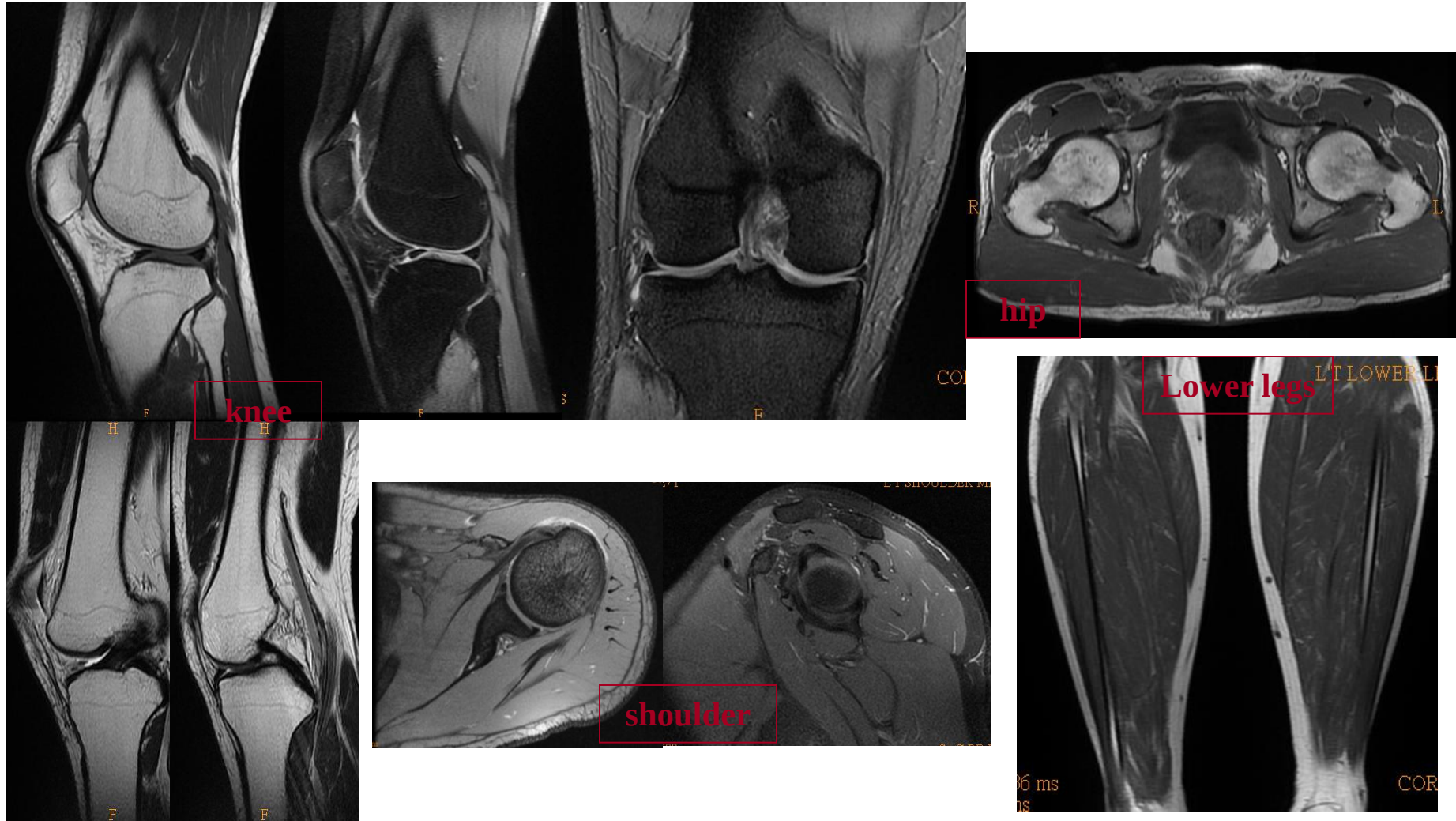
T1權重及脂肪抑制影像



STIR

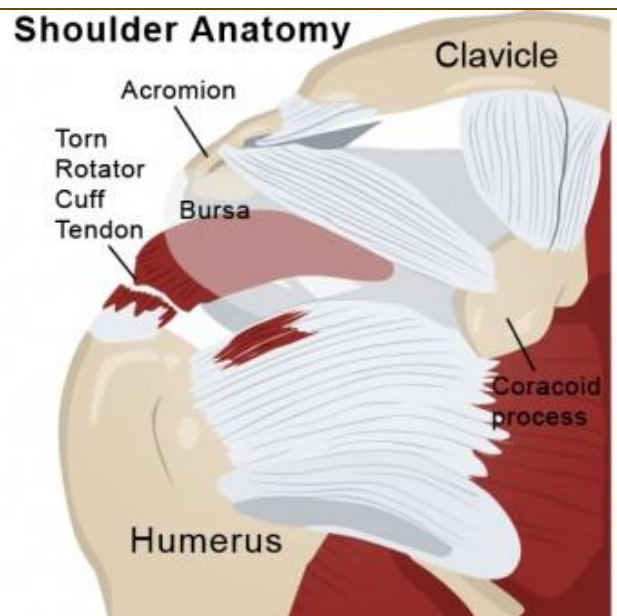
脂肪衰減影像

基礎磁共振造影及骨骼，軟骨，關節及軟組織解剖學知識

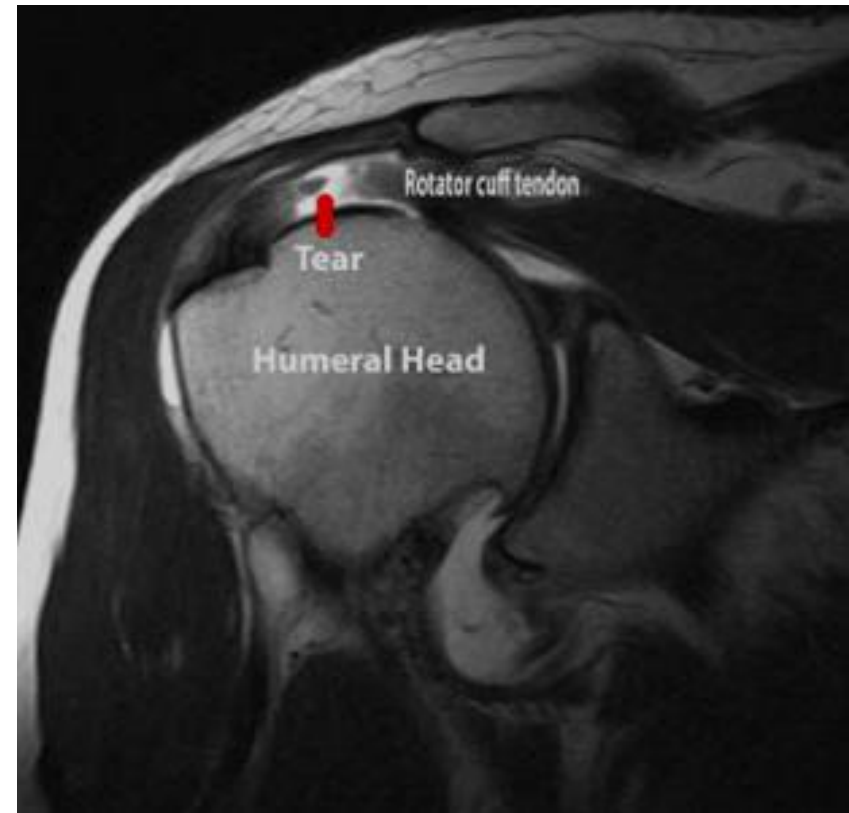
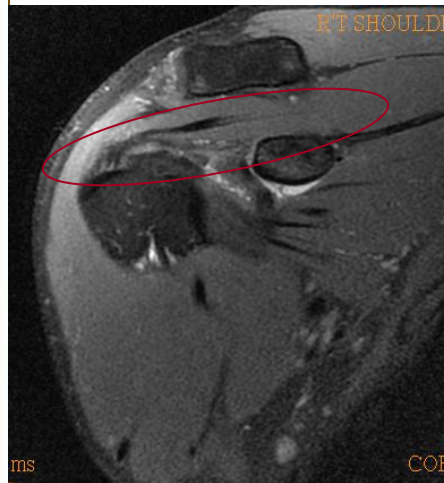


肩旋轉肌袖損傷 Rotator Cuff Injury

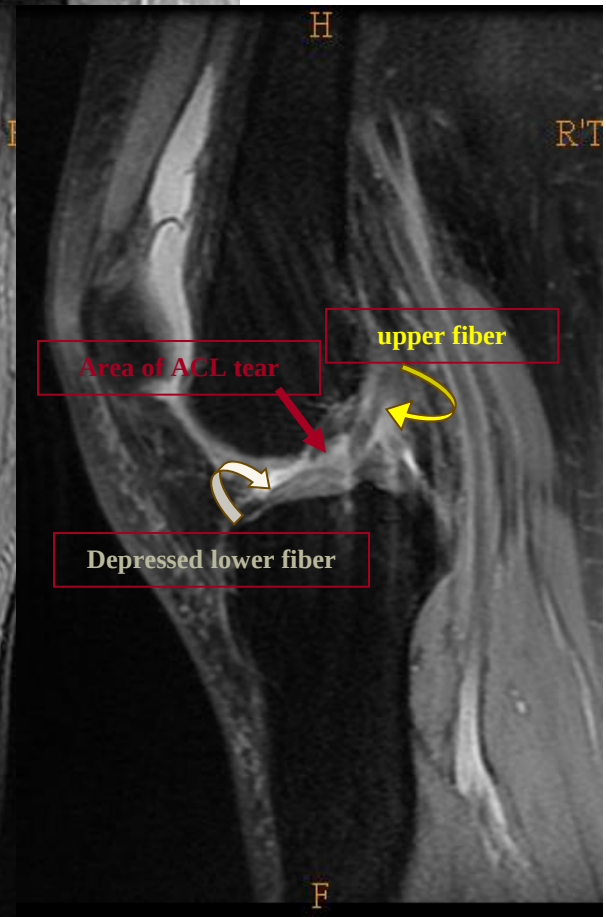
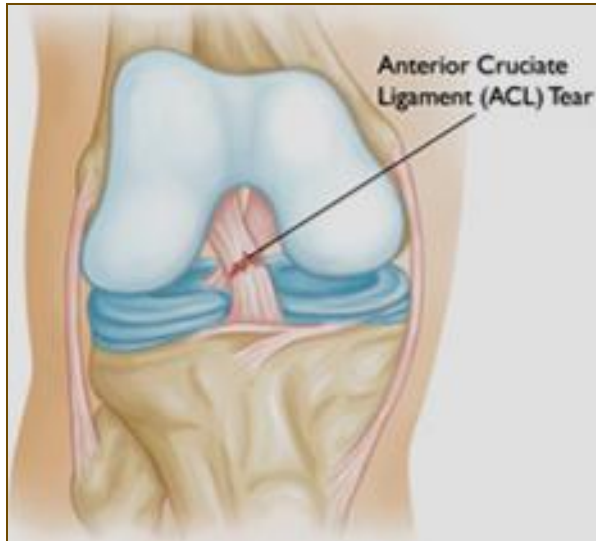
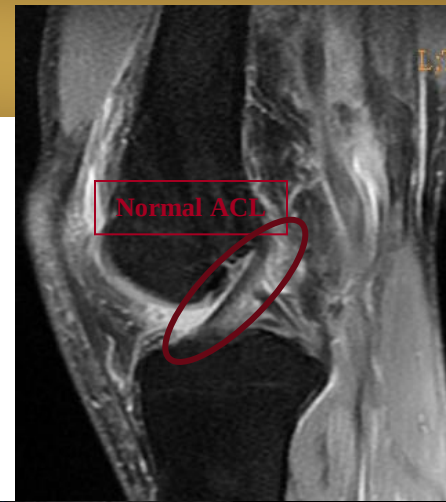
- Supraspinatus and infraspinatus tendons are more frequently torn
- Rotator cuff tears are very related to age, uncommon under age of 40



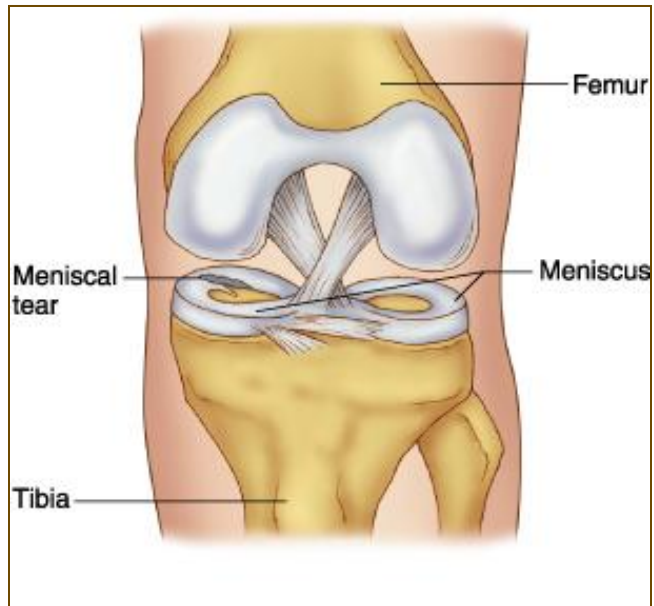
**Normal
supraspinatus tendon**



膝蓋前十字韌帶損傷

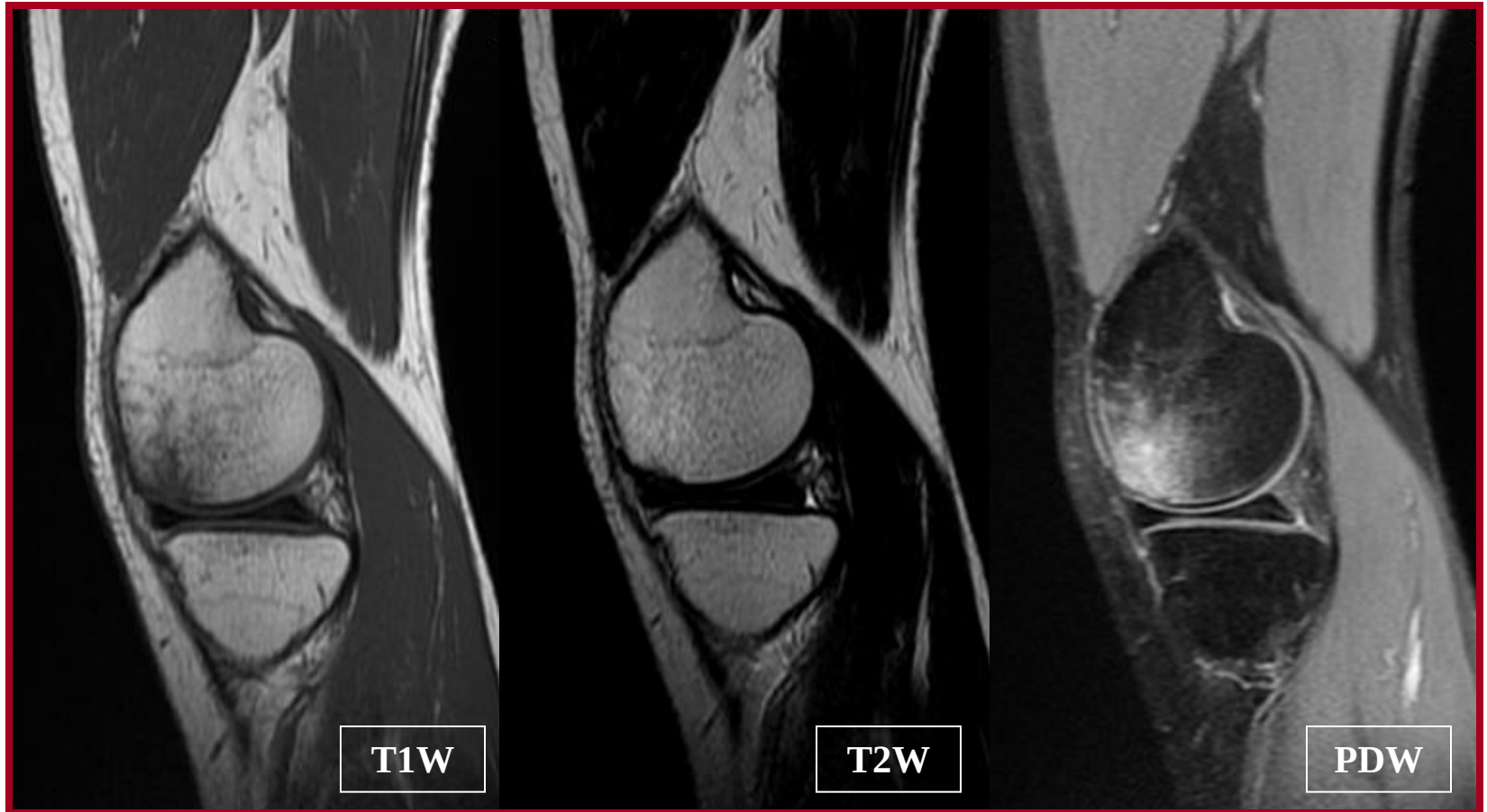


膝蓋半月軟骨損傷



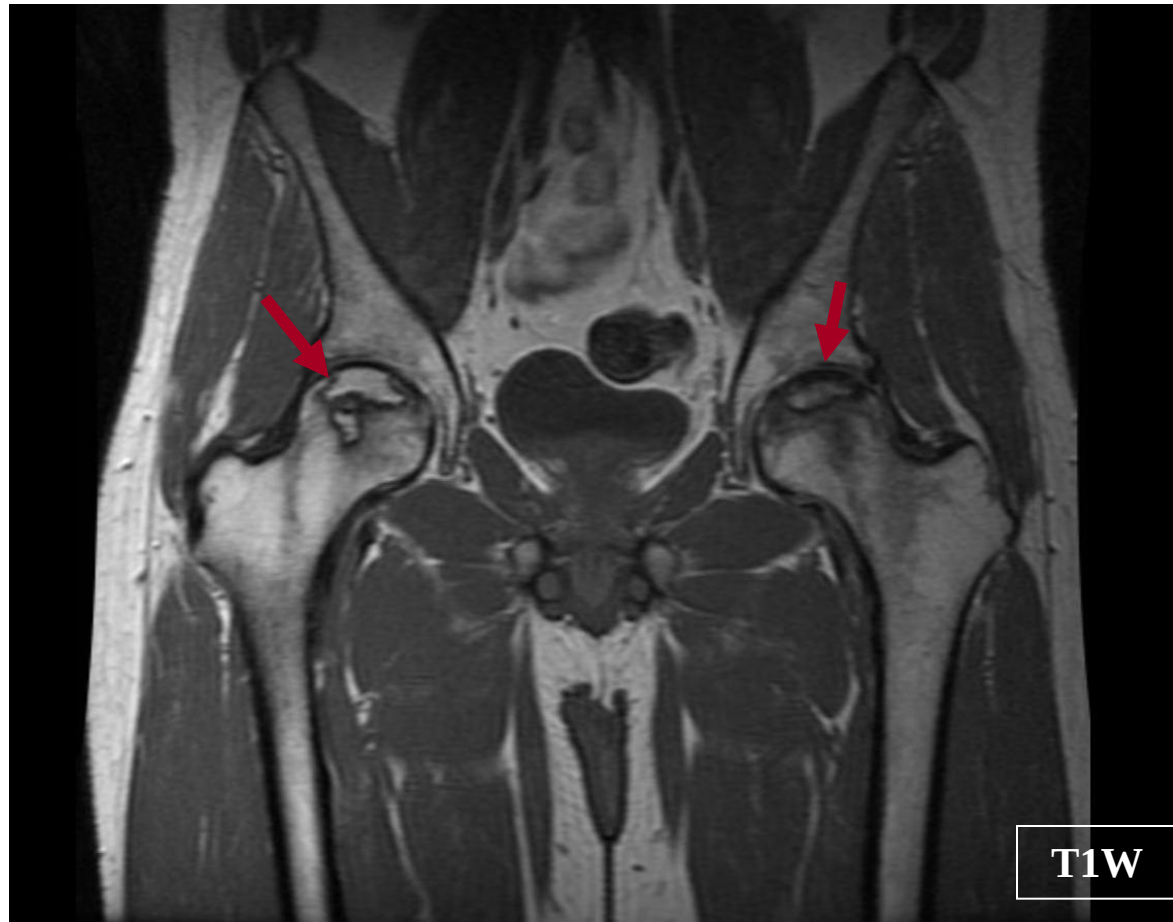
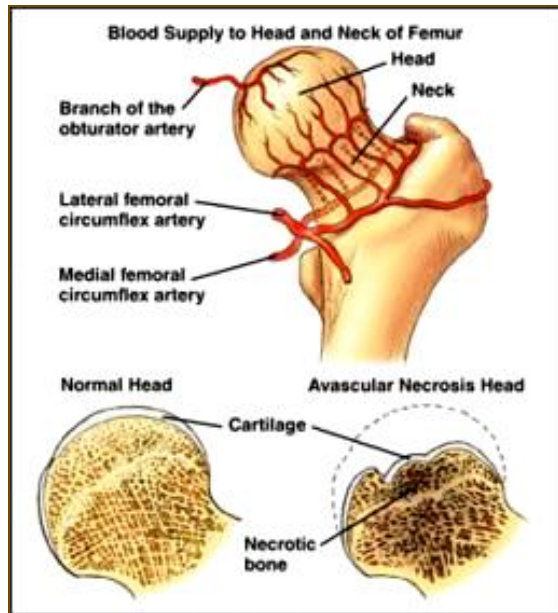
骨髓水腫 Bone marrow edema

- An inflammation that causes much than the needed fluid in a bone. It is normally caused by traumatic or non-traumatic conditions.



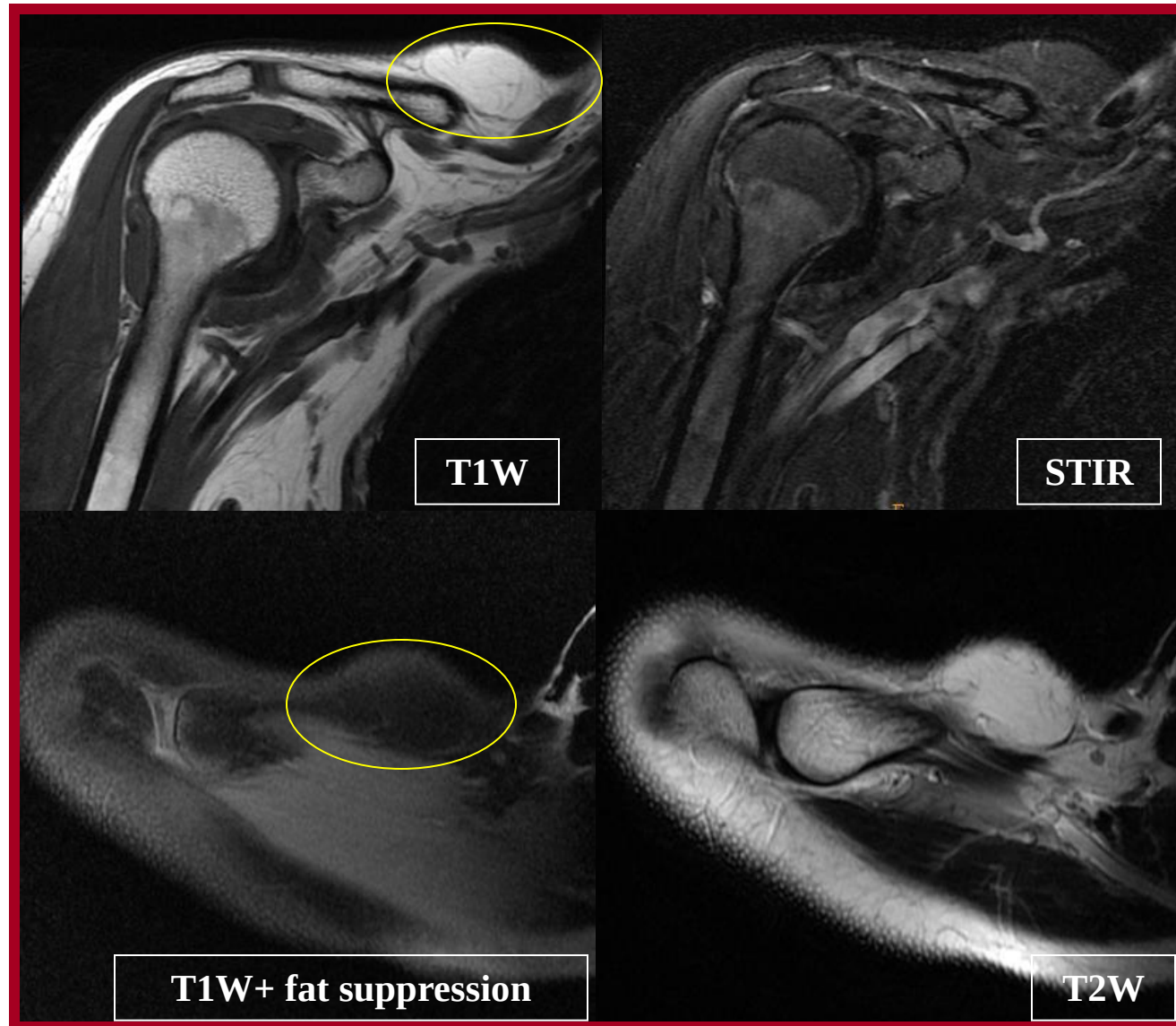
股骨頭缺血性壞死 Avascular necrosis of femoral Head

- Interruption of blood supply to the bone
- Conditions associated with alchhoholism, DM, extended periods of steroid usage, sickle cell disease, Gaucher's disease, trauma...



脂肪瘤

- Most common benign soft tissue tumor
- Subcutis
- Intramuscular
- Intermuscular
- Hyperintense on T1W, hypointensity on STIR images



神經瘤





Case demonstration on PACS

臺北醫學大學

TAIPEI MEDICAL UNIVERSITY



測驗

臺北醫學大學

TAIPEI MEDICAL UNIVERSITY

- 此為2014年1月14日
右膝磁共振造影影像
- 請問影像中箭頭所指為何異常？



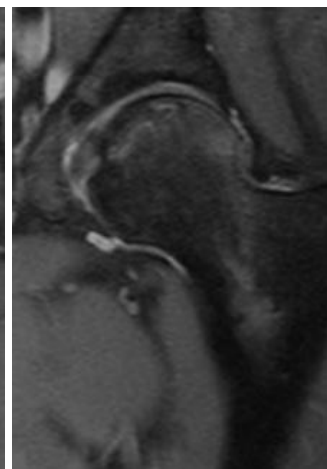
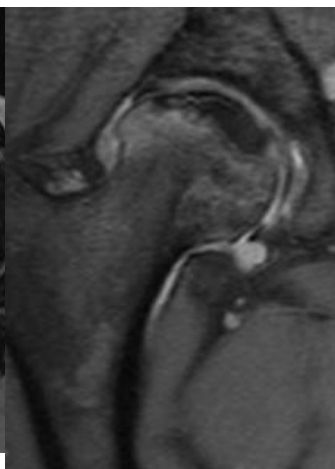
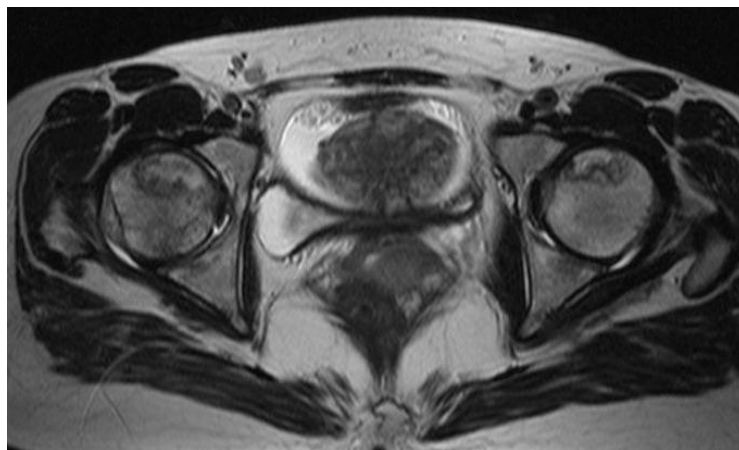
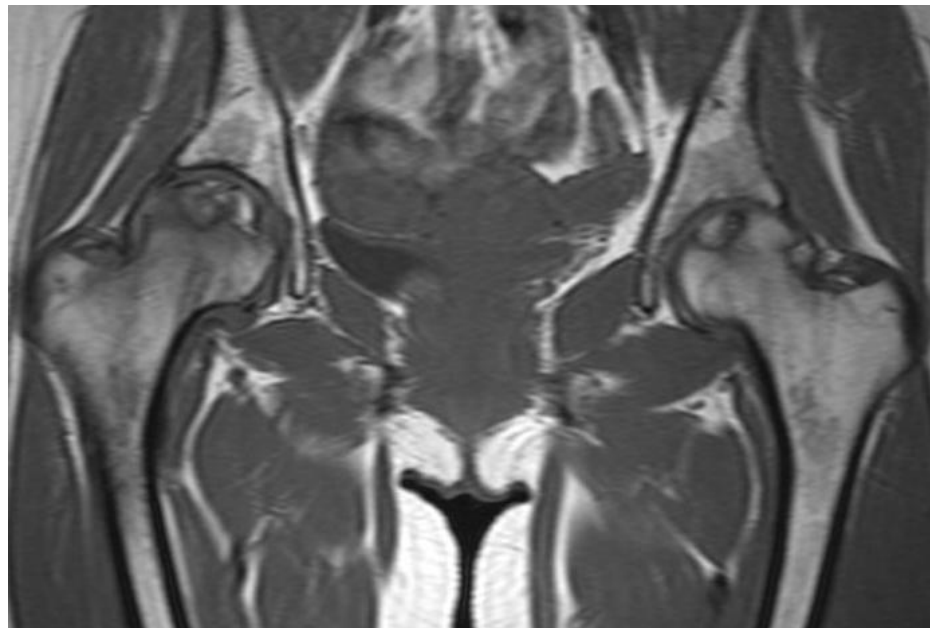
- 請問影像中紅色框線所指為何解剖結構？



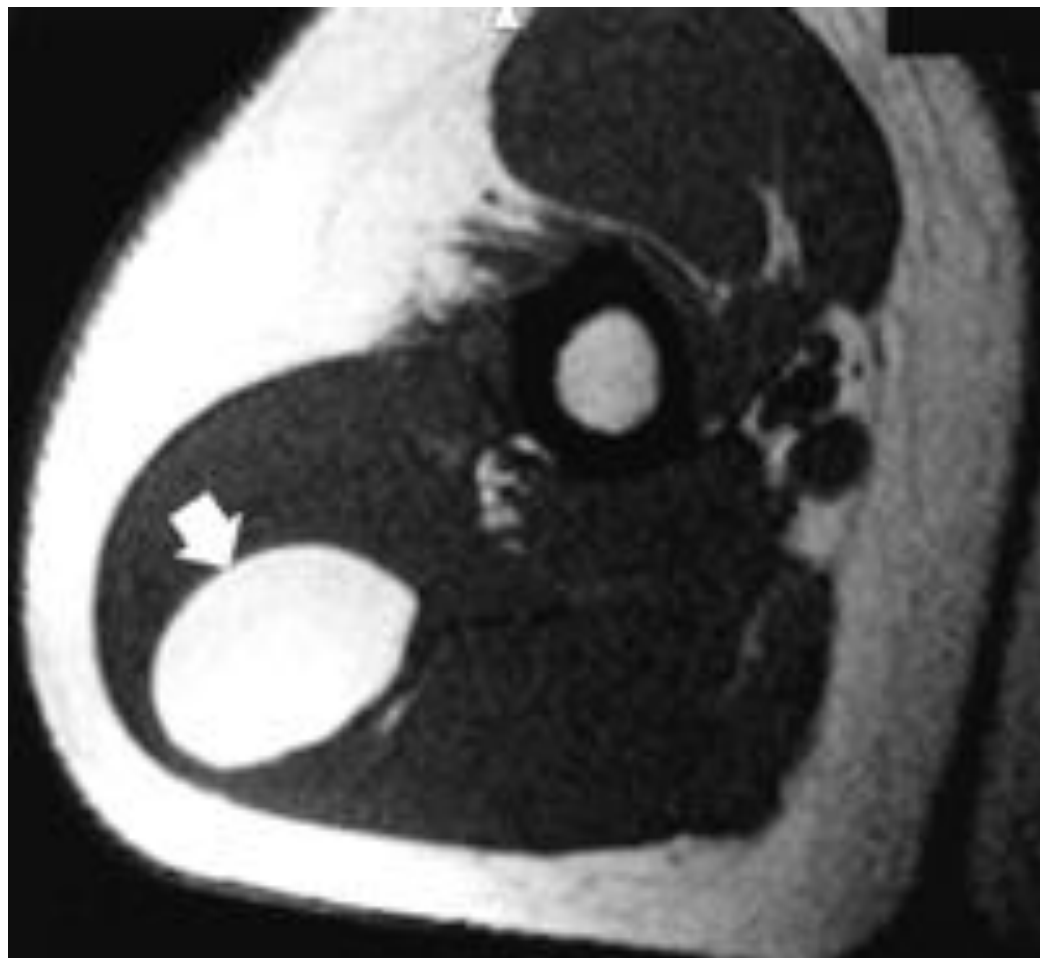
- 請問此為何種影像檢查序列？影像中箭頭所指為何解剖結構？



- 請問影像中箭頭所指為何種影像檢查序列？
- 影像中顯現何種病灶？
- 解剖位置為何？



- 請問此為何種影像檢查序列？
- 影像中顯現何種病灶？
- 診斷可能為何？



- 請問此為何種病灶？
- 注射對比劑後的影像中箭頭所指有何變化？

